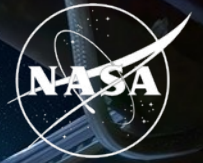


National Aeronautics and
Space Administration



Synthetic transcriptomics for mouse spaceflight

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EECS | UC Berkeley

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Biological & Physical Sciences

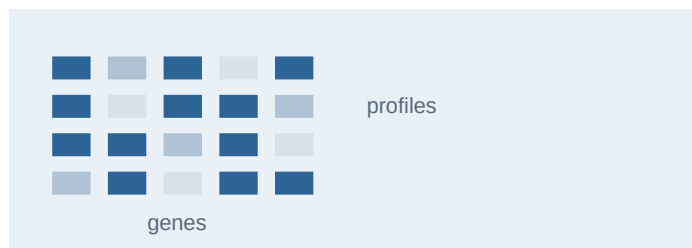


BACKGROUND

What is synthetic transcriptomics?

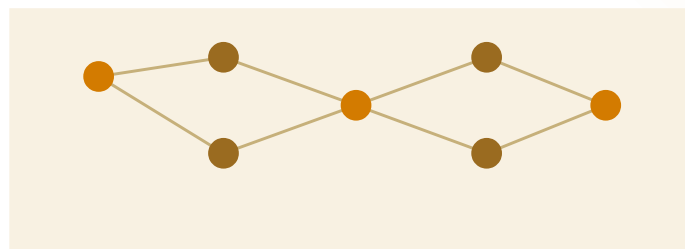
A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

01 Observed profiles



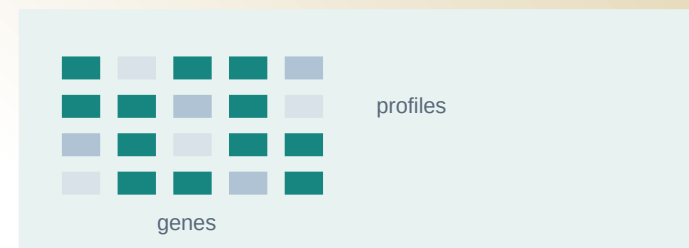
Measured gene expression from real mouse tissue

02 Learn the distribution



Capture gene relationships and variation by tissue and FLT/GC condition

03 Synthetic profiles



Sample new numeric expression vectors for a selected context

It can help

test classifiers, rank candidate genes and compare matched FLT and GC profiles

It does not add

new animals, new measurements or independent biological evidence

QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

Observed spaceflight cohort

1,610

profiles

75

accessions

835 FLT

775 GC



ARCHS4 mouse

Reference pretraining cohort

997,515

profiles screened



17,244

selected

20 tissues

Study effects can look like spaceflight biology.

- 01 Preserve tissue and FLT/GC structure.
- 02 Avoid memorizing individual profiles.
- 03 Test biological effects in observed OSDR samples.

STUDY GOALS

Match tissue distributions, then test FLT versus GC biology

The first goal validates the generator. The second asks whether synthetic data improves a real-data analysis.

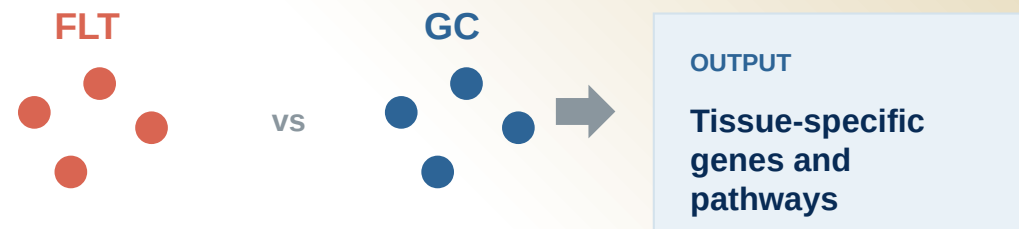
01 Reproduce tissue structure

Synthetic bulk RNA-seq should occupy the same tissue-defined expression space as real profiles.



02 Preserve the FLT/GC difference

Within each tissue, generated profiles should retain the smaller condition signal.



Model choice Use the pipeline that matches tissue structure and performs best on held-out real FLT/GC samples.

Gene effects and BH FDR use observed OSDR data.

METHOD

We built a configurable bulk RNA-seq generation pipeline

The downstream analysis used the outlined options; the remaining choices stay configurable.

01 Data scope

Sources

OSDR API

ARCHS4

Studies

Single

Multiple

Tissues

Per tissue

All tissues

02 Transform

Expression

Raw

CPM

TPM

Scaling

None

Z-score

MaxAbs

Features

All

HVG

L1000

03 Harmonization

Global or study correction

None

Within-study z-score

ComBat / MBatch

MOBER

04 Model training

Generator

WGAN-GP

DDIM

Training source

OSDR only

ARCHS4 only

Pretrain + adapt

05 Conditioning

Model inputs

Tissue

FLT / GC

Accession

Material type

Sex / age

Downstream

configuration

PREPROCESS

TPM / MaxAbs / 974 landmarks

PRETRAIN

ARCHS4

ADAPT

OSDR

GENERATE

Conditional DDIM

CONDITION

Tissue

FLT / GC

Accession

Material type

HARMONIZE

None

SCOPE

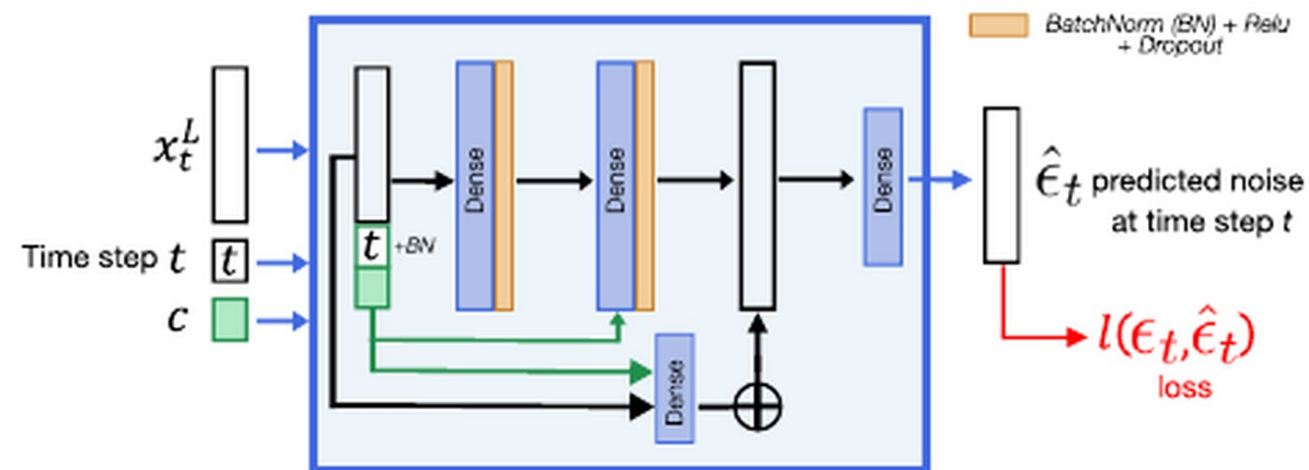
All tissues

VALIDATION

DDIM had lower separability and distributional distance

Both generators matched expression. DDIM was harder to distinguish from real profiles and had higher F1.

Architecture from Lacan et al.



Residual dense denoiser conditioned on timestep and tissue.

OSDR adaptation adds FLT/GC, accession and material context through LoRA.

ARCHS4 tissue probe

Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

Metric	WGAN-GP	DDIM	Read
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial acc.	0.636	0.475	near 0.5
FD / real P95	0.144	0.074	lower

Use DDIM

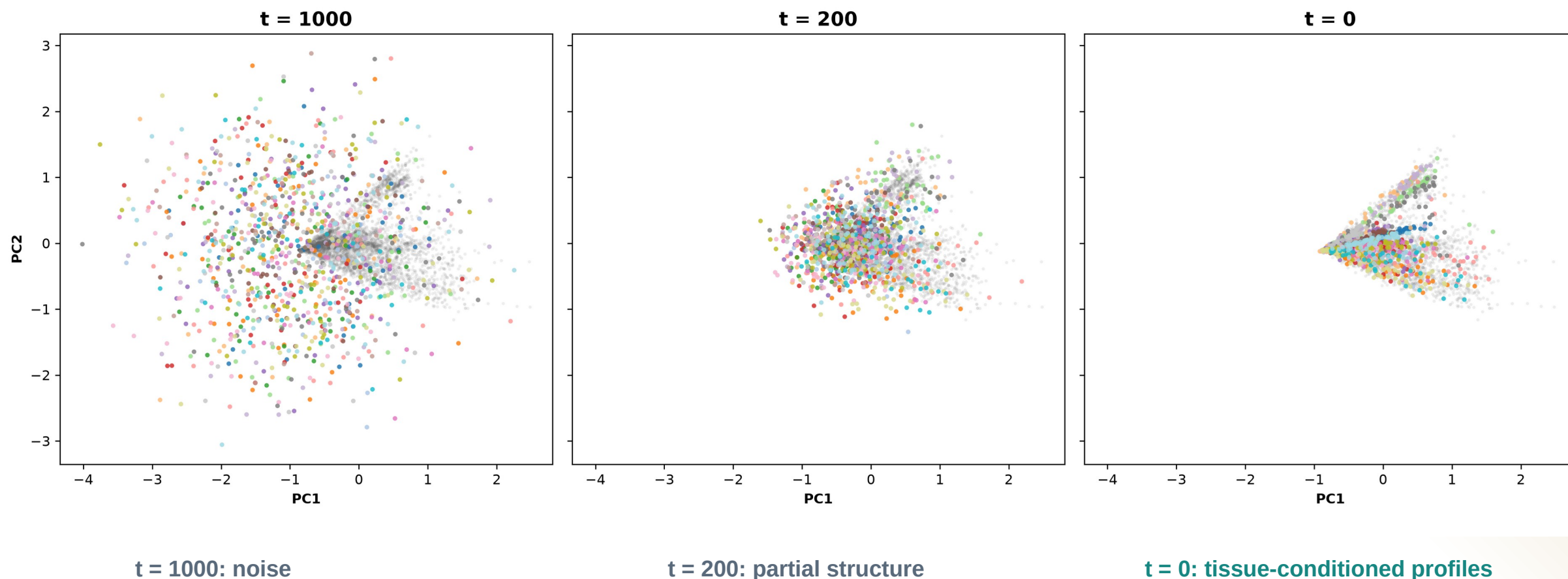
Higher F1 with lower separability and FD.

AA near 0.5 means real and generated profiles are difficult to separate.

DIFFUSION

Diffusion learns tissue structure from noise

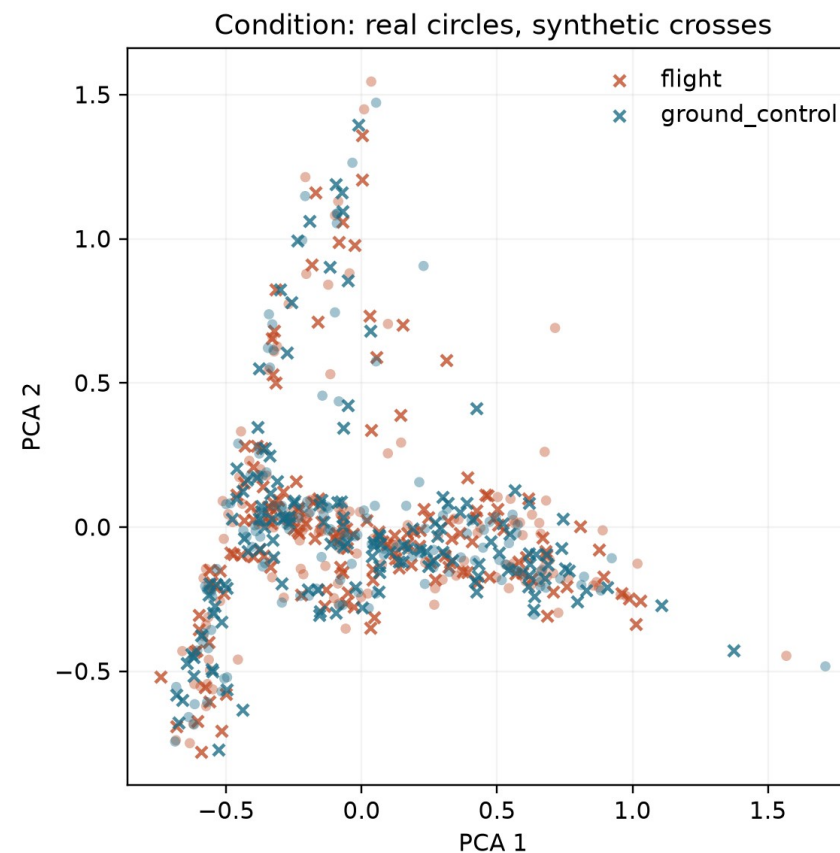
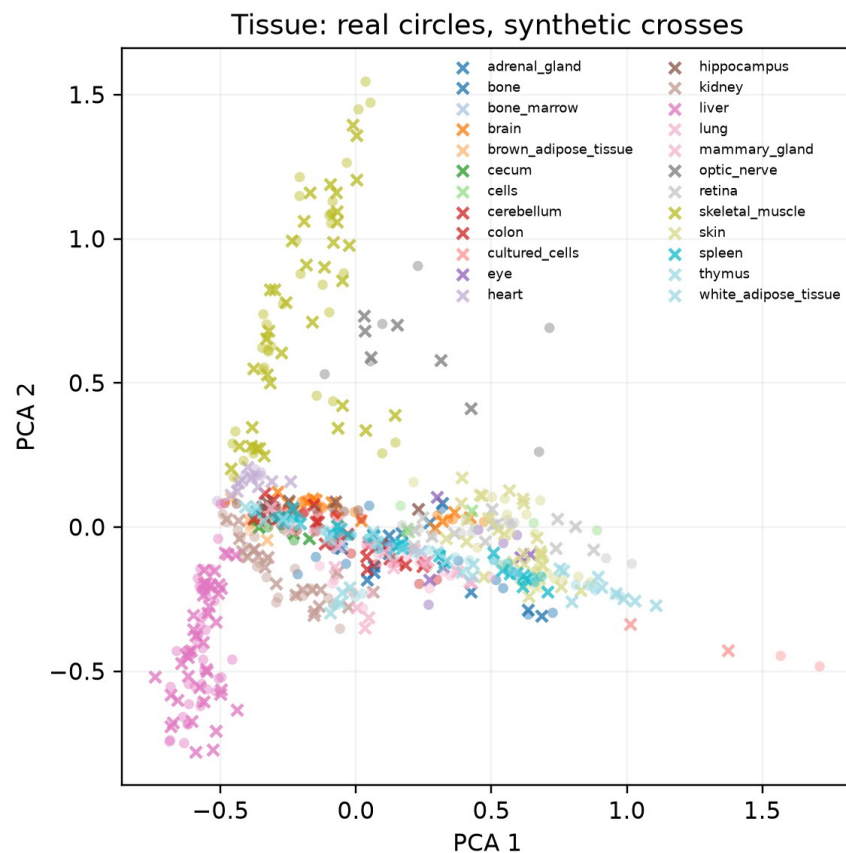
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



DIFFUSION OUTPUT

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.



Tissues separate clearly here. Flight and ground-control samples overlap much more, so we test that difference statistically.

Five arms separate gene ranking from classifier fitting

Guided arms use synthetic profiles to help choose genes; they do not treat generated profiles as new animals.

INPUTS		GUIDED		5%	S carries 5% of total fit weight	
ANALYSIS ARM		GENE RANKING		CLASSIFIER FIT		WHAT IT TESTS
Real only	R → Rank from real	R → Fit real	Baseline			
Generated only	S → Rank from generated	S → Fit generated	Synthetic-only stress test			
Real + generated	R + S → Consensus rank	R + S → Equal total weight	Direct augmentation			
Guided: real fit	R + S → Consensus rank	R → Fit real only	Feature guidance only			
Guided: 5% synthetic	R + S → Consensus rank	R + S → S = 5% weight	Guidance + light regularization			

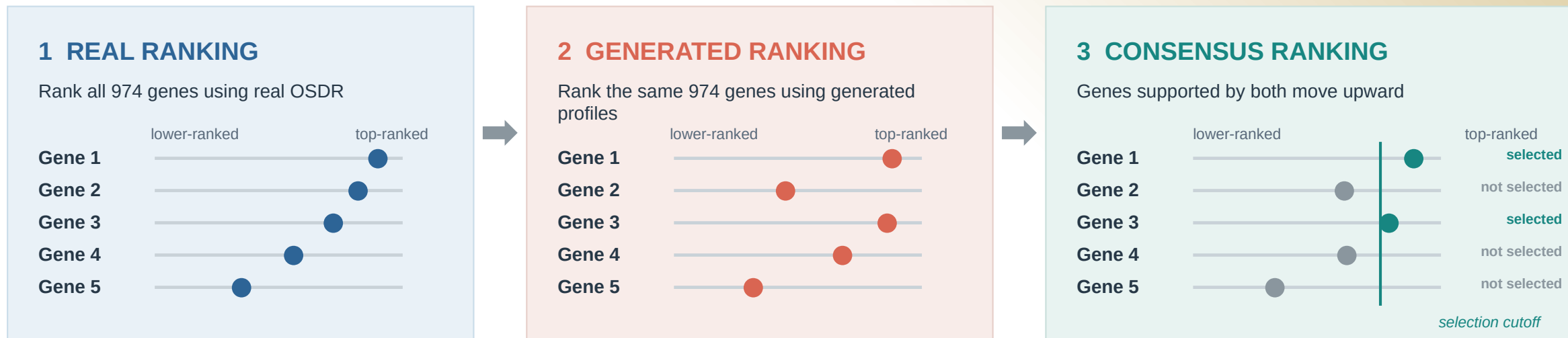
All five arms → Held-out real profiles → BA | AUROC | AP → Choose an eligible arm within each tissue

Association test FLT vs GC effects and BH FDR use real OSDR profiles only. Animal n stays unchanged.

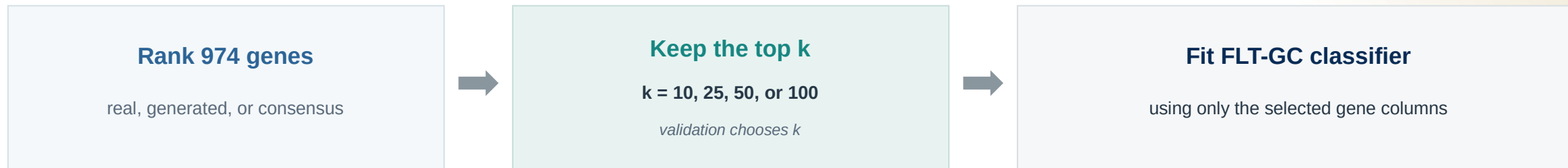
ANALYSIS

Consensus ranking chooses the classifier input genes

Real and generated rankings combine, then only the top k genes are used for FLT-GC classification.



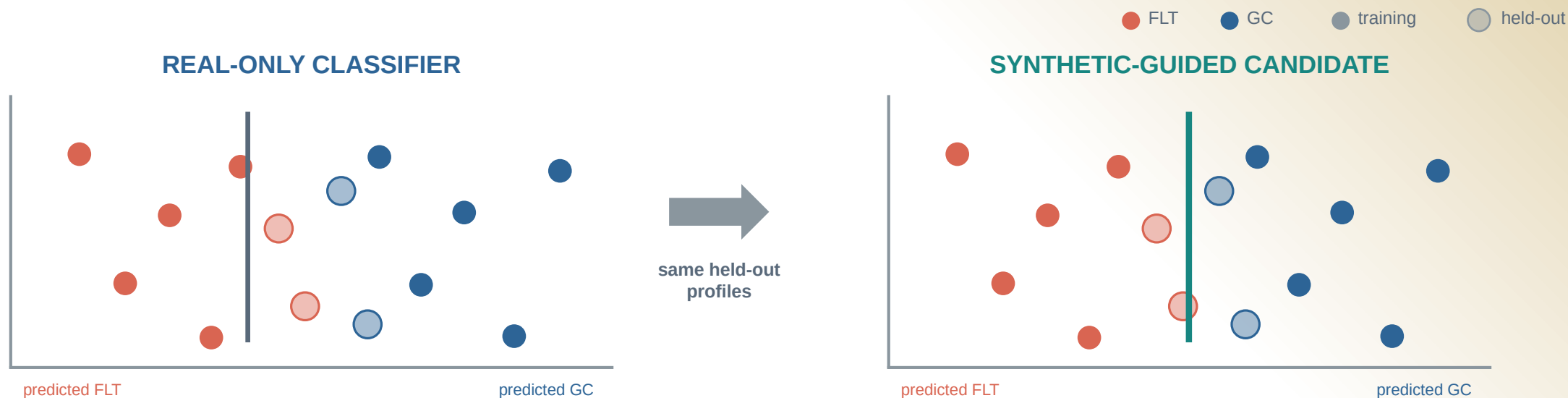
Top-ranked genes become classifier inputs



ANALYSIS

Held-out real samples decide whether guidance helps

Opaque profiles train the classifier; transparent profiles are used only to score generalization.



Compare on held-out real profiles

Balanced accuracy

AUROC

Average precision



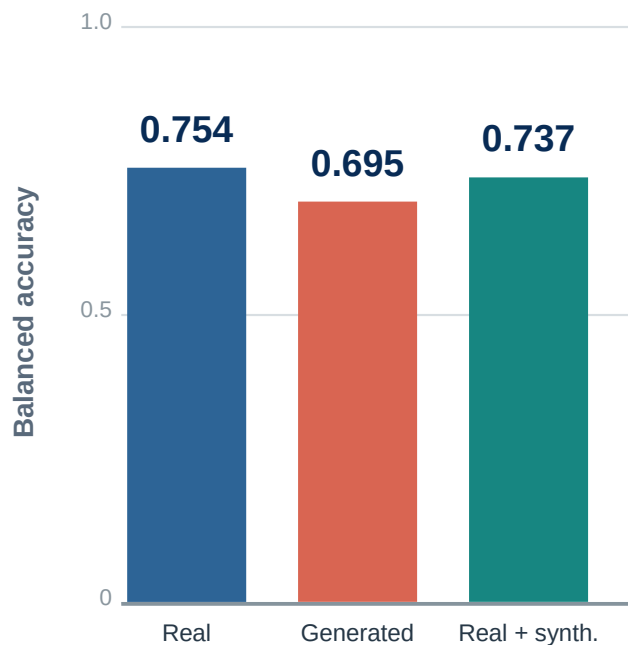
Retain the better eligible arm for that tissue

Held-out profiles never enter gene ranking, classifier fitting, or k selection.

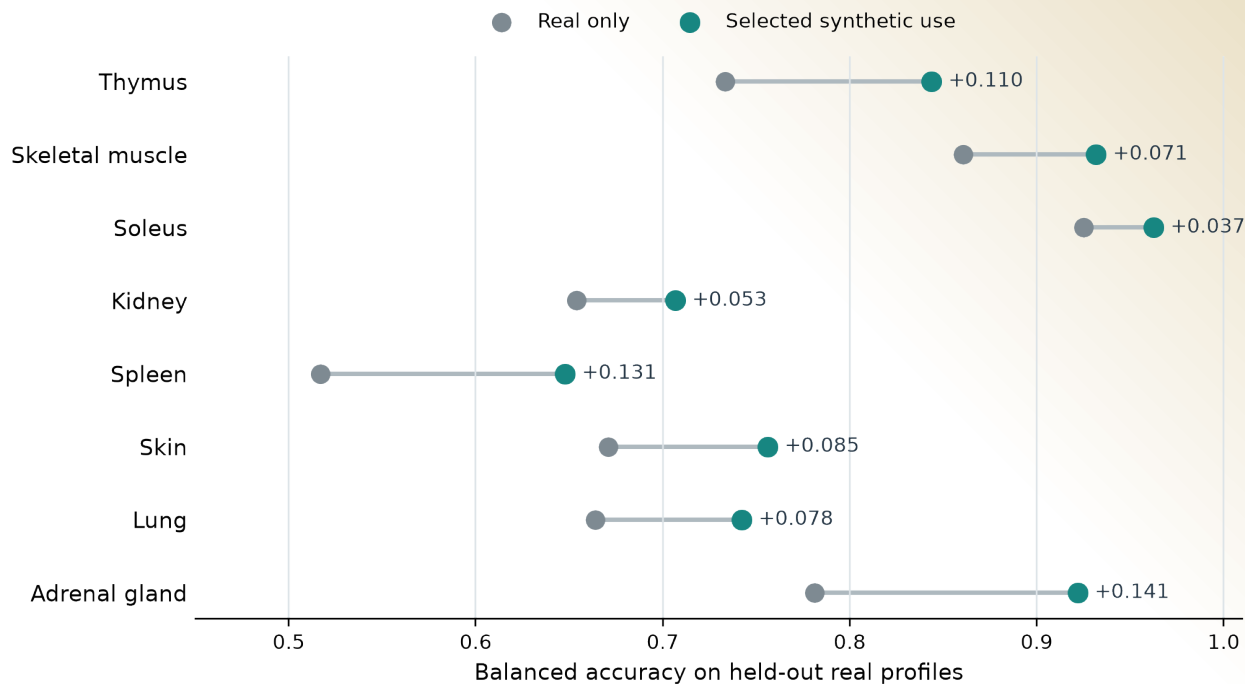
Pooled training missed improvements seen within tissues

The pooled FLT/GC classifier declined, while several tissue-specific classifiers improved.

Pooled across tissues



Selected use within each tissue



The useful arm differed by tissue.

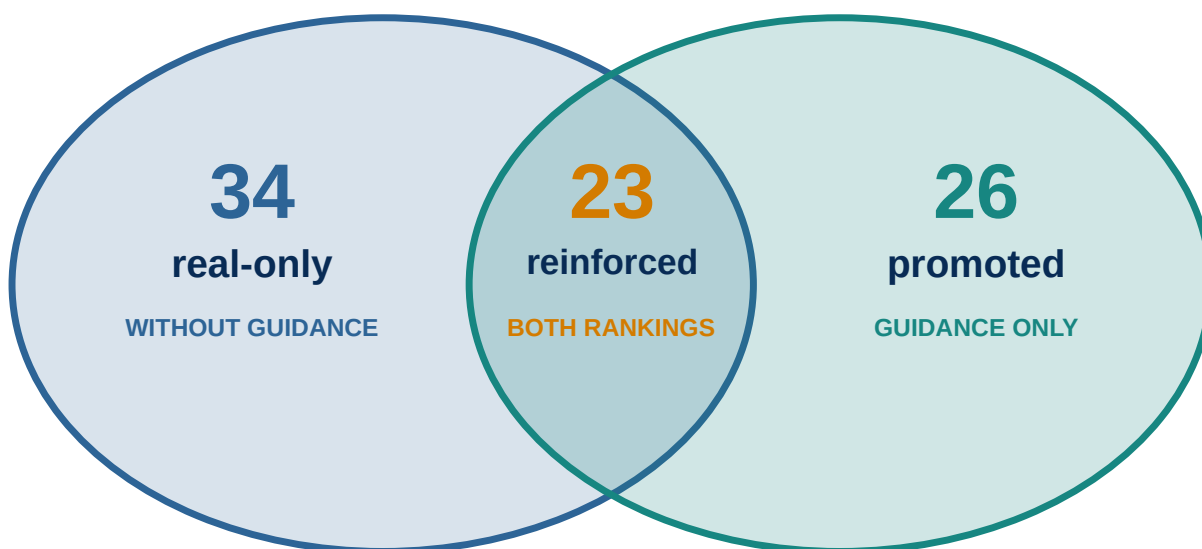
FEATURE ANALYSIS

Synthetic guidance changed ranking, not statistical evidence

Blue marks real-only selection; teal marks synthetic-guided selection. Every displayed association has real OSDR support.

Stable selection after real-data support

- Blue: stable in real-only ranking ● Teal: stable in synthetic-guided ranking



Real OSDR evidence gate

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

23 reinforced + 26 promoted
49 synthetic-informed associations

The 34 real-only associations remain real-data findings; synthetic guidance did not reinforce them.

LITERATURE INTERPRETATION

Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.

01 Fix the candidate

Gene, tissue, and FLT direction

02 Search

Spaceflight and mechanism literature

03 Compare

Observed result with published evidence

04 Record

Category, rationale, and source IDs

Selection status

26**Promoted**

Stable only with synthetic guidance

23**Reinforced**

Stable with and without guidance

Literature interpretation

Class	Count	Meaning
■ Aligning	22	Direct or same-tissue process agreement
■ Complementary	19	Related process or mechanism
■ Ambiguous	4	Mixed or context-dependent evidence
■ Unmatched	4	No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

ANALYSIS COVERAGE

The screen covered all 27 completed tissue analyses

The biological discussion narrows only after reporting synthetic-informed, real-only, and null outcomes.

10 Synthetic-informed

Promoted or reinforced BH-FDR gene

ANALYSIS UNITS

- Adrenal Gland
- Eye
- Kidney
- Skeletal muscle (pooled)
- Skin
- Spleen
- Thymus
- Gastrocnemius
- Soleus
- Tibialis Anterior

5 Real BH-FDR only

Real association, no synthetic-informed selection

ANALYSIS UNITS

- Heart
- Liver
- Retina
- EDL
- Quadriceps

12 No BH-FDR gene

No BH-FDR gene in the 974-gene panel

ANALYSIS UNITS

- Bone
- Bone Marrow
- Brain
- Brown Adipose Tissue
- Cecum
- Cerebellum
- Colon
- Hippocampus
- Lung
- Mammary Gland
- Optic Nerve
- White Adipose Tissue

GENE INVENTORY

Ten tissue analyses contained synthetic-informed genes

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

Rows: FLT direction + status

aligning

complementary

ambiguous

unmatched

Gene color = literature | Non-empty rows shown

Thymus 16 genes	FLT higher Promoted Hsd17b11, Etv1	
	FLT higher Reinforced Snx7	
	FLT lower Promoted Nusap1, Stmn1, Birc5, Cdk1, Top2a, Ccnb2, Aurka, Ccne2, Kif20a, Pcna, Ccnf	
	FLT lower Reinforced Ube2c, Gmnn	
Spleen 4 genes	FLT higher Promoted Rai14, Myl9, Ptprk	
	FLT higher Reinforced Loxl1	
Kidney 2 genes	FLT higher Promoted Inpp4b	
	FLT higher Reinforced Slc37a4	
Gastrocnemius 2 genes	FLT higher Promoted Nfkbia	
	FLT lower Promoted Fhl2	
Eye 1 gene	FLT lower Reinforced Klhl21	
Skeletal muscle (pooled) 12 genes	FLT higher Reinforced Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1	
	FLT lower Promoted Klhl21, Mapkapk5, Reep5, Itgb5	
	FLT lower Reinforced Bphl	
Soleus 5 genes	FLT higher Reinforced Tpm1	
	FLT lower Reinforced Bdh1, Ech1, Bnip3, Decr1	
Tibialis anterior 4 genes	FLT higher Promoted Cebpd	
	FLT higher Reinforced Cdkn1a, St3gal5, Bnip3	
Adrenal gland 2 genes	FLT lower Promoted Psmb8	
	FLT lower Reinforced Tspan4	
Skin 1 gene	FLT higher Promoted Plscr1	

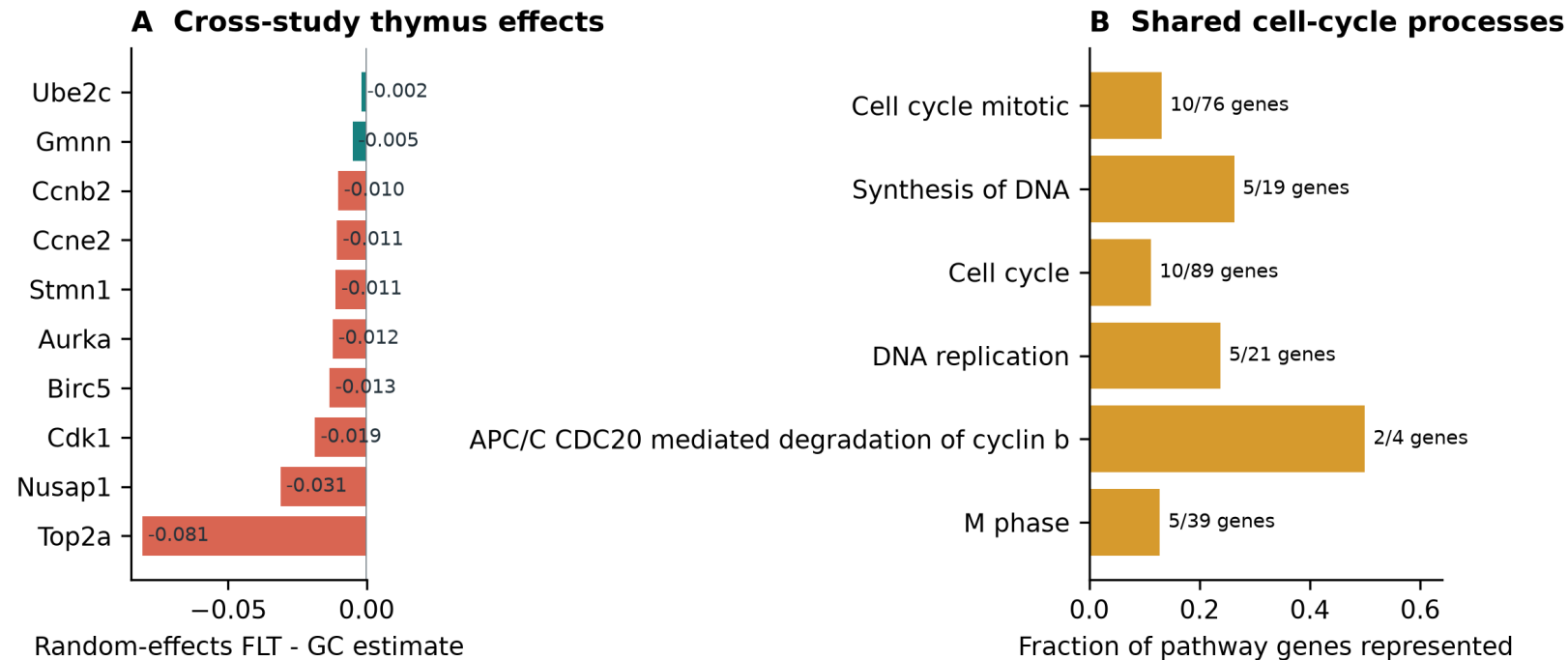
Separate rows report FLT direction and selection status; gene color independently reports literature interpretation for all 49 associations.

THYMUS

Thymus points to lower proliferative renewal

The cell-cycle program persisted without material conditioning; two FLT-higher genes extend the hypothesis.

Synthetic-informed thymus genes converge on a flight-lower mitotic program



16 synthetic-informed
BH-FDR associations

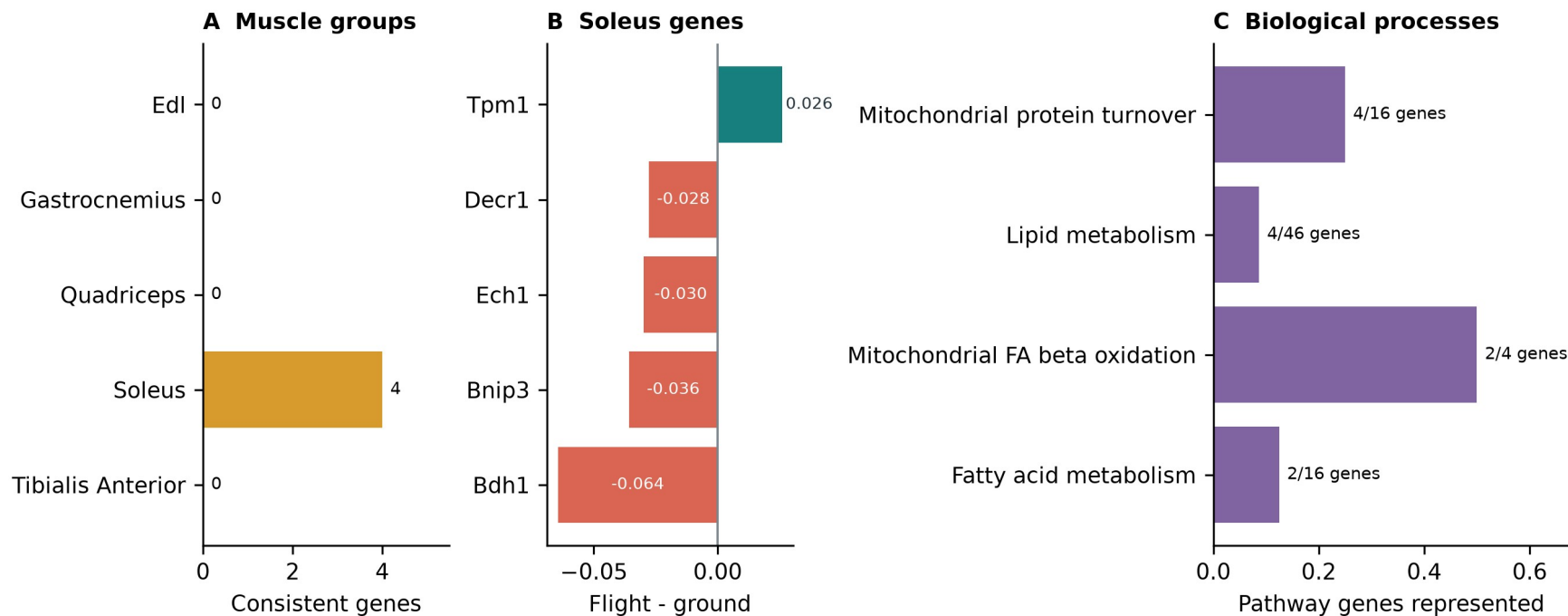
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Soleus reinforces a mitochondrial and lipid program

The real-data program is coherent, but its synthetic reinforcement required explicit material conditioning.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



0.925 to 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

ADDITIONAL FINDINGS I

Additional tissues produced distinct hypotheses

Promoted and reinforced selections are shown separately for each tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Pooled muscle	Promoted	Lower: Klhl21, Mapkapk5, Reep5, Itgb5	Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups.
	Reinforced	Higher: Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1; lower: Bphl	
Kidney	Promoted	Inpp4b higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
	Reinforced	Slc37a4 higher	
Spleen	Promoted	Rai14, Myl9, Ptprk higher	Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.
	Reinforced	Loxl1 higher	
Skin	Promoted	Plscr1 higher	Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.
	Reinforced	None	

ADDITIONAL FINDINGS II

Eye, adrenal and muscle-group results remain tissue-specific

Promoted and reinforced selections are separated within every tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Eye	Promoted	None	Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication.
	Reinforced	Klhl21 lower	
Adrenal gland	Promoted	Psmb8 lower	Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.
	Reinforced	Tspan4 lower	
Gastrocnemius	Promoted	Nfkb1a higher; Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
	Reinforced	None	
Tibialis anterior	Promoted	Cebpd higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.
	Reinforced	Cdkn1a, St3gal5, Bnip3 higher	

TAKEAWAYS

Synthetic data worked best as a tissue-specific prior

The generator helped prioritize observed signal. Biological claims still come from OSDR profiles.

01 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

02 Use

Tissue-specific consensus ranking and light synthetic regularization improved held-out prediction in selected tissues.

03 Interpret

Promoted and reinforced genes supported tissue-specific hypotheses. Literature review separated prior alignment from complementary findings.

Thank you

Questions?

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Mentor

James Casaletto

Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.